TECHNICAL DILIGENCE REVIEW - RECORD 016

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-016

1. Scope of the technical-debt review and its basis

This report, report TDR-2026-10, is the Wrenfield Technical Advisory LLP workstream on the technical-debt register, end-of-life dependency exposure and remediation follow-through, delivered to Ardent Peak on 2026-03-30. 2026-03-30 is the evidence cut-off for the present report, whose subject is the technical-debt register, end-of-life dependency exposure and remediation follow-through. The judgement the present report reaches is that the technical estate would support the business plan through the next twelve months and not through the fleet size the plan assumes for 2027 without the identified programme. Drawing the work together, TDR-2026-10 finds that the technical estate would support the business plan through the next twelve months and not through the fleet size the plan assumes for 2027 without the identified programme. J. Ferreiro notes that the company answered every document request, though two answers arrived after the draft.

Work on the present report ran over 7 weeks and included 10 interviews and read-only access to the engineering repositories. J. Ferreiro tested a sample rather than the population in TDR-2026-10, and the sampling basis is stated in each section. Two document requests remained partly unsatisfied when the report at TDR-2026-10 was delivered, and the affected conclusions are marked. TDR-2026-10 is limited by log retention: the telemetry window is ninety days, so events before 2026-01-05 could not be examined directly. The company reviewed a draft of the present report for factual accuracy and its corrections are reflected without altering any conclusion.

Grading in report TDR-2026-10 is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. The report at TDR-2026-10 grades evidence as strong, moderate or limited, and a limited grade is not upgraded because the consequence would be serious. Report TDR-2026-10 added two components to its sample after the first week, because the interview evidence pointed at them. For the present report the reviewer sampled 5 sites, including the largest deployment and the newest. J. Ferreiro has stated the limitations at the point they bite rather than collecting them into a disclaimer.

Nothing in TDR-2026-10 rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. The reviewer notes that report TDR-2026-10 was prepared while three remediation programmes were in flight, so several positions will have moved by completion. The interview programme behind report TDR-2026-10 covered 4 teams and produced 12 follow-up requests, of which most were satisfied. J. Ferreiro interviewed 10 engineers and managers for the present report; two declined to be quoted on one topic and that is noted where it matters. J. Ferreiro has not smoothed the sequence of this report; it follows the order in which the evidence arrived.

Selection for the present report favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. TDR-2026-10 sampled 19 services from an inventory of 34, weighted towards those on the critical control path. The costed remediation implied by this report is $1,505,000 over the coming four quarters, which is within the ordinary engineering budget. Deal implications in the report at TDR-2026-10 are modest in aggregate and specific in detail: 2 items for the closing checklist, 5 for the plan. Wrenfield Technical Advisory LLP has reported the position as at 2026-03-30 and has not attempted to predict where it will be at completion.

Interviews for this report were conducted with the owning engineer present rather than with a manager speaking on their behalf. The report at TDR-2026-10 draws on 10 interviews, including the engineering owners of every component examined in detail. Report TDR-2026-10 closes with 6 open items, listed with owner, evidence and the reviewer's view of whether the committed date is realistic. Three of the open items in this report would close on a single document that the company has agreed to produce. The numbering in the report at TDR-2026-10 follows the order of the work rather than the order of importance.

Schedule V - Costed technical-debt and migration register (TDR-2026-10)

Item	Area	Component	Effort estimate	Costed	Target date	Status
Partition the fleet allocator write path	scalability	Fleet Allocator	3.5 engineer-quarters	$420,000	2027-03-31	in design
Replace the unmaintained	IP and support	Localization Engine	1.5 engineer-quarters	$185,000	2026-12-31	committed

Item	Area	Component	Effort estimate	Costed	Target date	Status
scan-matching library in the localization fallback						
Retire the vendored geometry library in costmap fusion	support	Costmap Fusion	1 engineer-quarter	$128,000	2027-06-30	backlog
Raise robot software line coverage above sixty per cent on the perception path	quality	Costmap Fusion	2 engineer-quarters	$240,000	2027-03-31	in progress
Bring the second bench rig to the current sensor configuration	test capability	Hardware-in-the-Loop Rig	0.5 engineer-quarters	$96,000	2026-09-30	committed
Make the nightly simulation suite blocking	quality	Simulation Harness	1 engineer-quarter	$115,000	2026-11-30	blocked
Pin the full simulation build configuration and enable a determinism check	tooling	Simulation Harness	0.5 engineer-quarters	$68,000	2026-05-31	overdue
Replace the archived charting library in the operator console	security supportability	Remote Operations Console	1 engineer-quarter	$132,000	2027-01-31	backlog
Upgrade the streaming platform before its support window closes	support	Event Bus	1.5 engineer-quarters	$196,000	2027-06-30	committed
Migrate the deprecated warehouse transformation function	support	Analytics Warehouse Loader	0.5 engineer-quarters	$54,000	2026-12-31	committed
Re-converge the four forked site provisioning templates	configuration	Site Provisioning Service	0.5 engineer-quarters	$61,000	2026-09-30	in progress
Extend the edge buffer to twelve hours	resilience	Edge Gateway Agent	1 engineer-quarter	$144,000	2026-10-31	in progress
Replace the fallback scan matcher with the current implementation	performance	Localization Engine	1.5 engineer-quarters	$178,000	2026-12-31	in design
Introduce a second bench rig queue to cut firmware lead time	delivery	Hardware-in-the-Loop Rig	0.5 engineer-quarters	$88,000	2027-03-31	backlog
Add per-region credentials for the paging provider	security	Alarm Router	0.25 engineer-quarters	$34,000	2026-09-30	committed
Encrypt the on-robot localization map cache	security	Localization Engine	1 engineer-quarter	$126,000	2026-10-15	committed
Enforce a two-person rule on OTA artefact signing	security	OTA Updater	0.5 engineer-quarters	$72,000	2026-11-30	in design
Aggregate the safety controller fault log to the	safety evidence	Safety Controller	0.75 engineer-quarters	$104,000	2027-03-31	backlog

Item	Area	Component	Effort estimate	Costed	Target date	Status
cloud continuously						
Measure closing rate for the separation monitor	safety	Speed and Separation Monitor	2 engineer-quarters	$268,000	2026-08-31	in progress
Validate and ship the wheel slip detector	safety	Wheel Odometry Filter	1.5 engineer-quarters	$162,000	2026-09-15	in progress
Refit the battery state-of-health estimator on opportunity-charge data	quality	Battery Management Service	1 engineer-quarter	$138,000	2026-09-30	in progress
Move the calibration record for the payload scale to the cloud	quality	Payload Scale Interface	0.25 engineer-quarters	$42,000	2027-01-31	backlog
Split the shared database instance used by the allocator and the charge scheduler	reliability	Charge Scheduler	0.75 engineer-quarters	$92,000	2026-12-31	committed
Publish the modified files carrying file-level copyleft obligations	IP hygiene	Costmap Fusion	0.25 engineer-quarters	$28,000	2026-09-30	committed

2. The costed register: TD-23 and the funded items

In the programme TD-23 on Charge Scheduler is behind a supplier release outside the company's control. Charge Scheduler carries register item TD-23, the work to split the shared database instance used by the allocator and the charge scheduler. On funding, TD-23 is funded from the second quarter onward. register item TD-23 would be a candidate for the first post-close engineering review under any reasonable plan.

Costmap Fusion item TD-03 is the item to retire the vendored geometry library in costmap fusion, classified under support. Programme item TD-03 carries an estimate of 1 engineer-quarter; the costed figure is $128,000. TD-03 carries this observation: management ranked this behind the separation monitoring work. TD-03 on Costmap Fusion is included because a reader of TDR-2026-10 should be able to disagree with the inference without repeating the work.

Programme item TD-07 is sized at 0.5 engineer-quarters, or $68,000 at the company's loaded rate. Reading the programme plan, register item TD-07 is last in a three-item chain, so its date is the least reliable on the register. Under tooling, entry TD-07 is to pin the full simulation build configuration and enable a determinism check. Simulation Harness item TD-07 is the reason one section of TDR-2026-10 concludes narrowly rather than generally.

Register item TD-12 is unfunded and dependent on capital rather than engineering effort in the engineering plan the reviewer was shown. Edge Gateway Agent item TD-12 is in progress with a target date of 2026-10-31. In the programme TD-12 is behind a supplier release outside the company's control. Edge Gateway Agent item TD-12 has no single accountable owner, and that is the finding the reviewer would press hardest.

Security item TD-16 is partly funded, with the second phase unbudgeted; the register status and the funding position are not the same thing. What TD-16 protects against is the fact that it a stolen compute module discloses customer warehouse layouts. Localization Engine item TD-16 carries a 2026-10-15 target and a status of committed. register item TD-16 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Effort on programme item TD-02 is estimated at 1.5 engineer-quarters, costed at $185,000. The plan treats Localization Engine item TD-02 as funded by reallocating effort from feature work, which the reviewer checked against the headcount model. The IP and support entry at Localization Engine item TD-02 is to replace the unmaintained scan-matching library in the localization fallback. TD-02 on Localization Engine was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Programme item TD-15 is sized at 0.25 engineer-quarters, or $34,000 at the company's loaded rate. As at the review date programme item TD-15 was committed, with 2026-09-30 still the stated target. Unfunded and dependent on capital rather than engineering effort is the funding position on TD-15. Alarm Router item TD-15 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Funded and already staffed is the funding position on programme item TD-09. Progress on TD-09 is recorded as committed against a 2027-06-30 date. The support entry at programme item TD-09 is to upgrade the streaming platform before its support window closes. TD-09 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-10.

As at the review date TD-13 on Localization Engine was in design, with 2026-12-31 still the stated target. Localization Engine item TD-13 matters because it relocalisation exceeds the operator prompt timeout at sparse sites. The dependency structure places TD-13 on Localization Engine behind a customer maintenance window that is not yet agreed. performance item TD-13 was checked against the corresponding entry in the company's own register, and the register was current.

The test capability entry at register item TD-05 is to bring the second bench rig to the current sensor configuration. The consequence of leaving Hardware-in-the-Loop Rig item TD-05 undone is that it one sensor path is verified on a single rig. About register item TD-05 it should be said that outstanding since the April 2025 lidar timing incident. programme item TD-05 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

As at the review date register item TD-04 was in progress, with 2027-03-31 still the stated target. Funding for quality item TD-04 is not funded, and carried in the register regardless, and the register does not say so. Sequencing of TD-04 matters: it is behind a customer maintenance window that is not yet agreed. TD-04 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

For Analytics Warehouse Loader item TD-10 the team estimates 0.5 engineer-quarters, which the reviewer converted to $54,000 for comparison. Programme item TD-10 is committed with a target date of 2026-12-31. The reviewer's qualification on TD-10 is that the warehouse vendor has published a withdrawal date. programme item TD-10 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Schedule T - Third-party component and licence inventory (TDR-2026-10)

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
nav-graph-solver	2.4.1	Apache-2.0	no	linked into the planner as a static library	approved 2024-02-19	Motion Planner
geom2d-fusion	0.9.7	LGPL-2.1	weak	dynamically linked by the fusion pipeline	approved 2023-08-04	Costmap Fusion
scanmatch-legacy	1.2.0	GPL-2.0	strong	linked into the localization fallback path	not approved	Localization Engine
grpc-transport	1.62.0	Apache-2.0	no	service-to-service transport across the cloud	approved 2024-05-30	Event Bus
tsdb-client	3.1.4	MIT	no	telemetry writes from the ingest service	approved 2024-09-11	Telemetry Ingest
chartkit	4.8.2	BSD-3-Clause	no	operator console visualisations	approved 2023-11-27	Remote Operations Console
physics-core-fork	0.14.3	Zlib	no	simulation physics	approved with conditions 2023-04-18	Simulation Harness
rtos-shim	1.0.9	BSD-2-Clause	no	real-time helper layer on the robot	approved 2024-01-23	Motion Planner
protobuf-runtime	25.3	BSD-3-Clause	no	message encoding across edge and cloud	approved 2024-04-08	Config Distribution
sqlite-embedded	3.45.1	public domain dedication	no	on-robot local state	approved 2023-06-15	Edge Gateway Agent

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
opencv-subset	4.9.0	Apache-2.0	no	image preprocessing for grasp proposals	approved 2024-07-02	Gripper Controller
pointcloud-utils	2.1.6	MPL-2.0	weak	point cloud filtering in fusion	approved 2024-03-14	Costmap Fusion
yaml-parse	0.15.2	MIT	no	configuration parsing on the robot	approved 2023-09-29	Config Distribution
jwt-verify	5.2.1	MIT	no	token verification in the identity service	approved 2024-06-21	Identity and Authorization
kafka-client	3.6.1	Apache-2.0	no	event bus producer and consumer clients	approved 2024-02-27	Event Bus
media-transcode	6.0.0	LGPL-2.1 with an optional GPL build	weak	video snippet transcoding	approved with conditions 2024-08-16	Video Snippet Service
wms-adapter-sdk	2.2.0	commercial, per-seat	no	one customer's warehouse system adapter	approved 2023-12-05	WMS Connector
lidar-vendor-sdk	4.1.2	commercial, redistributable binary	no	lidar driver	approved 2023-05-22	Lidar Driver
safety-controller-toolchain	3.0.4	commercial, development use only	no	safety controller build toolchain	approved 2023-02-10	Safety Controller
cell-model-lib	1.8.0	Apache-2.0	no	battery state estimation	approved 2024-10-01	Battery Management Service
grid-render	3.3.5	GPL-3.0 with a commercial alternative	strong	map authoring geometry rendering	approved with conditions 2023-07-19	Map Authoring Studio
http-router	1.9.4	MIT	no	cloud service routing	approved 2024-01-09	Order Broker
otel-instrument	1.24.0	Apache-2.0	no	tracing across cloud services	approved 2024-05-14	Fleet Health Monitor
pgbouncer-managed	1.21.0	ISC	no	database connection pooling	approved 2024-03-05	Fleet Allocator
wheel-slip-model	0.3.1	AGPL-3.0	strong	prototype slip detector, not shipped	not approved	Wheel Odometry Filter
csv-export	2.0.3	MIT	no	billing and usage exports	approved 2023-10-17	Billing Usage Meter
pager-provider-sdk	3.4.0	commercial, subscription	no	alarm dispatch	approved 2023-03-28	Alarm Router
k8s-operator-framework	1.31.0	Apache-2.0	no	cloud deployment controllers	approved 2024-09-24	Site Provisioning Service
image-signing	2.5.0	Apache-2.0	no	artefact signature verification	approved 2025-01-15	OTA Updater
warehouse-sim-assets	1.1.0	CC BY-SA 4.0	strong	simulation scenario assets	approved with conditions 2024-11-06	Simulation Harness

3. End-of-life dependencies and supportability

The estimate for programme item TD-21 is 1 engineer-quarter and $138,000. Left in place, the condition behind TD-21 cell imbalance can hide until derating. Reading TD-21 against the funded plan, data collection is running at the two affected sites. register item TD-21 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

For TD-19 on Speed and Separation Monitor the team estimates 2 engineer-quarters, which the reviewer converted to $268,000 for comparison. The reviewer's comment on register item TD-19 is that the most significant open safety item in the reviewed record. Safety item TD-19 is in progress with a target date of 2026-08-31. register item TD-19 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-10.

1.5 engineer-quarters is the estimate on safety item TD-20, which converts to $162,000. TD-20 on Wheel Odometry Filter appears in the register and is funded in the current financial year. TD-20 on Wheel Odometry Filter is in progress with a target date of 2026-09-15. TD-20 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The reviewer's qualification on quality item TD-06 is that blocked on the physics core determinism work. Programme item TD-06 is sized at 1 engineer-quarter, or $115,000 at the company's loaded rate. TD-06 on Simulation Harness is not funded, and carried in the register regardless in the engineering plan the reviewer was shown. TD-06 on Simulation Harness was inspected in the repository rather than described from the ticket, which is why the account is specific.

The reviewer's comment on Safety Controller item TD-18 is that the safety team supports it; it is not funded in the current plan. Aggregate the safety controller fault log to the cloud continuously is entered at safety evidence item TD-18 as an safety evidence item. What TD-18 on Safety Controller protects against is the fact that it investigation evidence is lost between weekly extractions. safety evidence item TD-18 was checked against the company's own register, which had not been updated since the change landed.

Effort on register item TD-08 is estimated at 1 engineer-quarter, costed at $132,000. 2027-01-31 is the target for TD-08 on Remote Operations Console, whose status is backlog. Programme item TD-08 is funded from the second quarter onward; the register status and the funding position are not the same thing. TD-08 on Remote Operations Console was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

On funding, TD-11 is not funded, and carried in the register regardless. The configuration entry at configuration item TD-11 is to re-converge the four forked site provisioning templates. What TD-11 protects against is the fact that it template drift produced a go-live failure in May 2026. programme item TD-11 is the clearest instance in TDR-2026-10 of a control that exists on paper and is weaker in practice.

Sizing on register item TD-22 gives 0.25 engineer-quarters and $42,000. Item TD-22 on Payload Scale Interface in the debt register is to move the calibration record for the payload scale to the cloud, and it bears on Payload Scale Interface. TD-22 is sequenced ahead of two other items that depend on it. TD-22 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Register item TD-24 carries a 2026-09-30 target and a status of committed. The business consequence recorded against TD-24 is that it three modified files under a file-level copyleft licence appear unpublished. On funding, Costmap Fusion item TD-24 is funded in the current financial year. register item TD-24 is the reason one section of TDR-2026-10 concludes narrowly rather than generally.

About register item TD-14 it should be said that requires capital for a third rig rather than engineering effort. 2027-03-31 is the target for register item TD-14, whose status is backlog. TD-14 is the item to introduce a second bench rig queue to cut firmware lead time, classified under delivery. programme item TD-14 has no single accountable owner, and that is the finding the reviewer would press hardest.

The plan treats Fleet Allocator item TD-01 as funded from the second quarter onward, which the reviewer checked against the headcount model. Programme item TD-01 carries a 2027-03-31 target and a status of in design. If Fleet Allocator item TD-01 is not done, the exposure is that it caps the region at roughly eight sites at current order density. register item TD-01 is reported here in full because a summary of it in an earlier draft of TDR-2026-10 was misread as an assurance.

Funding for register item TD-17 is unfunded and dependent on capital rather than engineering effort, and the register does not say so. OTA Updater item TD-17 carries this observation: the control is designed but the signing tool does not enforce it. The dependency structure places TD-17 on OTA Updater ahead of two other items that depend on it. OTA Updater item TD-17 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Schedule K - Penetration-test findings by severity with residual risk (TDR-2026-10)

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
SEC-2025-01	High	7.4	Identity and Authorization	2025-09-16	2025-11-14	low once lifetimes are cut to one

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
						hour with refresh
SEC-2025-02	High	8.1	Release Orchestrator	2025-09-17	2025-09-19	low; redaction is enforced in the logging library
SEC-2025-03	Medium	6.2	Remote Operations Console	2025-09-18	2026-01-22	low; pinning is applied at the gateway and the agent
SEC-2025-04	Medium	5.4	Map Authoring Studio	2025-09-18	2025-11-27	medium; the shortened publication path is still outstanding
SEC-2025-05	High	7.8	Telemetry Ingest	2025-09-19	2026-02-06	low; attestation is now required on the ingest path
SEC-2025-06	Medium	6.5	Video Snippet Service	2025-09-19	2025-12-18	low; the index is scoped per tenant and the identifiers are opaque
SEC-2025-07	Medium	5.9	Site Provisioning Service	2025-09-22	2026-01-16	medium; two forked templates were not re-issued until May 2026
SEC-2025-08	Medium	6.0	Event Bus	2025-09-23	2025-10-31	low; the interface is restricted to the operations subnet
SEC-2025-09	Medium	6.3	OTA Updater	2025-09-24	pending	medium; a two-person rule is designed but not enforced in the tool
SEC-2025-10	Low	3.7	Charge Scheduler	2025-09-24	2025-10-09	low
SEC-2025-11	Low	3.1	Remote Operations Console	2025-09-25	2025-10-30	low
SEC-2025-12	Low	4.2	Analytics Warehouse Loader	2025-09-25	not planned	medium; management accepted the risk with a compensating audit trail
SEC-2026-01	High	7.1	Release Orchestrator	2026-04-14	committed 2026-08-31	high until the second-approver control lands
SEC-2026-02	High	7.6	Robot Shadow Store	2026-04-15	2026-05-29	low; authorization is checked on both the cache and durable paths
SEC-2026-03	High	8.4	Config Distribution	2026-04-15	2026-06-12	low; the flag was removed and the agent refuses unsigned bundles
SEC-2026-04	Medium	5.6	WMS Connector	2026-04-16	2026-05-20	low; payload logging requires an explicit per-tenant approval
SEC-2026-05	Medium	5.2	Alarm Router	2026-04-16	committed 2026-09-30	medium; per-region credentials are scheduled but not issued
SEC-2026-06	Medium	6.1	Simulation Harness	2026-04-17	pending	medium; the role was scoped down but still reads one production bucket
SEC-2026-07	Medium	6.7	Hardware-in-the-Loop Rig	2026-04-17	committed 2026-10-31	medium; network segmentation is the only current control
SEC-2026-08	Medium	6.4	Identity and Authorization	2026-04-18	2026-06-30	low; role grants now require a second approver

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
						and are alerted
SEC-2026-09	Low	3.9	Fleet Health Monitor	2026-04-18	2026-05-08	low
SEC-2026-10	Low	4.0	Billing Usage Meter	2026-04-19	2026-06-05	low; object lock is applied and verified nightly
SEC-2026-11	Low	4.4	Gripper Controller	2026-04-19	committed 2026-11-30	medium; the safety controller limit is the compensating control
SEC-2026-12	Low	3.5	Edge Gateway Agent	2026-04-20	pending	medium; a wipe procedure exists but was not applied retroactively
SEC-2026-13	Low	3.3	Order Broker	2026-04-20	2026-05-15	low
SEC-2026-14	Low	3.8	Traffic Manager	2026-04-21	2026-06-18	low
SEC-2026-15	Low	3.0	Site Provisioning Service	2026-04-21	not planned	low; management judged the operational cost of enforcement higher than the risk
SEC-2026-16	Medium	5.8	Costmap Fusion	2026-04-22	2026-06-26	low; the socket binds to the loopback interface only
SEC-2026-17	High	7.9	Motion Planner	2026-04-22	2026-06-19	low; the port requires a signed local credential
SEC-2026-18	Low	3.4	Battery Management Service	2026-04-23	committed 2026-12-15	low; no fleet impact demonstrated
SEC-2026-19	Medium	5.1	Localization Engine	2026-04-23	committed 2026-10-15	medium; disk encryption is planned for the next compute revision
SEC-2026-20	Low	3.2	Payload Scale Interface	2026-04-24	not planned	low; management accepted pending the calibration service redesign

4. The services carrying the largest debt

Ownership of Fleet Health Monitor sits with Field Reliability; Wanda Njoku signs off changes to it. Fleet Health Monitor pairs a Python 3.11 implementation with cloud, containers on a managed Kubernetes cluster, and its data lives in a time-series store plus PostgreSQL 14 for case records. The published objective for Fleet Health Monitor is that a fleet-affecting condition raises an alert inside three minutes of first symptom. The cadence recorded for Fleet Health Monitor is fortnightly, which matches the observed release history. Fleet Health Monitor was weighed on its coverage, its defect backlog and its dependency position, and the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Fleet Health Monitor is the reason one section of TDR-2026-10 concludes narrowly rather than generally.

One coupling concern applies to Fleet Health Monitor: a time-series store plus PostgreSQL 14 for case records is shared with a second service, so a schema change is not a local decision. Deployment of Fleet Health Monitor is cloud, regional, which sets the blast radius of any defect in it. Field Reliability tracks 17 open defects on Fleet Health Monitor, including the acknowledged case where alert grouping collapses distinct faults at the same site into one case, hiding the second fault. Inputs to Fleet Health Monitor come from the telemetry ingest path and the robot shadow store, and its consumers are the alarm router and the field service workflow. Test coverage on Fleet Health Monitor is 57 per cent line coverage and the measurement basis was confirmed with Wanda Njoku. Fleet Health Monitor was raised again with the owning engineer after the initial interview, and the position did not change.

Against Order Broker, the coverage figure is 78 per cent line coverage, which the figure is respectable but excludes the fallback path, which is where the known defect sits. Fleet Services tracks 11 open defects on Order Broker, including the acknowledged case where order cancellation races with assignment and can leave an orphaned task record. Data ownership is clear for Order Broker: it writes PostgreSQL 14 with an event journal and no other service writes there. For Order Broker the supportability position is none identified. Order Broker was assessed against the objective and the defect record, and the design is sound and the test evidence is thinner than the design deserves. Order Broker was excluded from an earlier draft of TDR-2026-10 and reinstated after the reviewer obtained the supporting record.

For Order Broker the stated target is that no accepted order is lost, measured by journal reconciliation each night, expressed against production traffic rather than a benchmark. Order Broker sits cloud, regional, and no alternative placement is currently supported. Upstream of Order Broker sit the WMS connector and the site configuration service; downstream, the fleet allocator and the analytics warehouse loader consume its output. Order Broker releases weekly. The evidence base for Order Broker was the broker journal schema, the nightly reconciliation output and two cancellation incidents. Order Broker has no single accountable owner, and that is the finding the reviewer would press hardest.

Alarm Router sits at 51 per cent line coverage; the trend across the reviewed period was up by seven points. Field Reliability publishes one objective for Alarm Router: a dispatched alarm reaches an on-call responder inside 90 s at p95. Changes to Alarm Router reach production fortnightly. Alarm Router was weighed on its coverage, its defect backlog and its dependency position, and the component is adequate today and carries a supportability liability that is not on the funded plan. The paging provider and the customer notification channel rely on Alarm Router, which in turn relies on the fleet health monitor and the on-call rotation source. Alarm Router was checked against the corresponding entry in the company's own register, and the register was current.

Responsibility for Alarm Router was confirmed with Wanda Njoku, whose team, Field Reliability, also holds the on-call duty. The evidence base for Alarm Router was the routing table export, delivery receipts for one quarter and two missed-page reviews. End-of-life review of Alarm Router found the paging provider's older application interface is still in use. Data ownership is clear for Alarm Router: it writes PostgreSQL 14 for routes and delivery receipts and no other service writes there. Alarm Router carries 14 open defects, among them a case where an on-call rotation gap in the routing table falls back to a shared mailbox nobody watches at night. Alarm Router is one of the entries where TDR-2026-10 relies on the company's own measurement rather than an independent one.

The dependency map places the customer directory federations and the secret store above Identity and Authorization and every cloud service and the remote operations console below it. Of the 6 open items against Identity and Authorization, the reviewer treats one as material: service-to-service tokens are issued with a twelve-hour lifetime, longer than the design intent. The assessment of Identity and Authorization is not unqualified: the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. What the reviewer had for Identity and Authorization was the authorization model document, a token lifetime audit and the secret rotation log; the conclusions below do not reach past it. Implementation of Identity and Authorization rests on Go 1.22; the process runs on cloud, containers on a managed Kubernetes cluster and persists to PostgreSQL 14 with a hardware-backed key store. Identity and Authorization would be a candidate for the first post-close engineering review under any reasonable plan.

Against Identity and Authorization, the coverage figure is 84 per cent line coverage, which coverage alone overstates the assurance here, because the bench suite is the real safety net. One coupling concern applies to Identity and Authorization: PostgreSQL 14 with a hardware-backed key store is shared with a second service, so a schema change is not a local decision. The cadence recorded for Identity and Authorization is weekly, with a security review on every change, which matches the observed release history. Identity and Authorization is deployed cloud, global control plane with regional caches. Identity and Authorization was tested against the dependency ageing register, which records none identified in the reviewed manifests. Identity and Authorization is reported here in full because a summary of it in an earlier draft of TDR-2026-10 was misread as an assurance.

The motion planner and the remote operations console rely on Traffic Manager, which in turn relies on the fleet allocator and the site map service. The cadence recorded for Traffic Manager is weekly, paired with the allocator release, which matches the observed release history. For Traffic Manager the stated target is that corridor deadlock detected and broken inside two seconds, expressed against production traffic rather than a benchmark. Implementation of Traffic Manager rests on Go 1.22; the process runs on cloud, containers on a managed Kubernetes cluster and persists to Redis 7

for corridor reservations with a PostgreSQL 14 audit trail. Traffic Manager is owned by the Fleet Services team, with Owen Brantley named as the engineering owner of record. Traffic Manager is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The backlog for Traffic Manager is 12 items deep and contains an acknowledged defect whereby a corridor reservation leaks when a robot drops off the network mid-traverse. 74 per cent line coverage is reported for Traffic Manager, and the reviewer re-ran the measurement on a clean checkout. The assessment of Traffic Manager is not unqualified: no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. The evidence base for Traffic Manager was the reservation protocol specification and a replay of three deadlock events. In production Traffic Manager is cloud, regional, with a degraded on-robot yield behaviour if the link drops, unchanged since the last architecture review. Traffic Manager rests on a single source, and the reviewer has said so rather than implying corroboration.

58 per cent line coverage is reported for Costmap Fusion, and the reviewer re-ran the measurement on a clean checkout. The reviewer's judgement on Costmap Fusion is that the component would not block the transaction, and it would appear on any competent post-close engineering plan. Reading the dependency map from Costmap Fusion outward gives the lidar driver, the depth camera bridge and the wheel odometry filter on the input side and the motion planner and the speed and separation monitor on the output side. The objective recorded against Costmap Fusion is that fused occupancy grid published at 20 Hz with sensor-to-grid latency under 60 ms, and it is reported weekly to the engineering leadership. Evidence examined for Costmap Fusion comprised the fusion design note, sensor timing captures and the reviewer's own bench replay. Costmap Fusion has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Implementation of Costmap Fusion rests on C++17; the process runs on on-robot Linux, pinned to the drive control core and persists to a shared-memory ring buffer with no durable persistence. Costmap Fusion's dependency ageing was checked: a vendored geometry library whose upstream branch left support in 2024. Of the 21 open items against Costmap Fusion, the reviewer treats one as material: a stale lidar frame is retained for one extra cycle while a driver reconnects. Costmap Fusion releases monthly, batched with perception model updates. Deployment of Costmap Fusion is edge, co-scheduled with the planner on the compute module, which sets the blast radius of any defect in it. Costmap Fusion was checked against the corresponding entry in the company's own register, and the register was current.

Gripper Controller carries a single measurable objective, namely that a grasp completes or reports failure inside 1.2 s. Robot Firmware releases Gripper Controller quarterly, and no release in the sampled window bypassed the gate set. Gripper Controller has a single named owner, Ines Duarte of Robot Firmware, which is not true of every service reviewed. Coverage for Gripper Controller was 41 per cent line coverage at the review date, and the trend over the reviewed period was up by seven points. Supportability of Gripper Controller turns on one fact: none; the coprocessor part is current production. Gripper Controller was raised again with the owning engineer after the initial interview, and the position did not change.

Behind Gripper Controller sits a local parameter store; the service itself is C++17 on on-robot Linux with a firmware coprocessor. Gripper Controller sits edge, only on the two robot variants fitted with a manipulator, and no alternative placement is currently supported. Of the 15 open items against Gripper Controller, the reviewer treats one as material: force limiting relies on a calibration that is only re-run when a gripper is replaced. Inputs to Gripper Controller come from the payload scale interface and the perception grasp proposals, and its consumers are the order execution path and the safety controller fault input. For Gripper Controller the reviewer worked from the gripper control specification, calibration records for eleven units and a force bench test. Gripper Controller is one of the entries where TDR-2026-10 relies on the company's own measurement rather than an independent one.

Payload Scale Interface carries 11 open defects, among them a case where the calibration record is stored locally and is lost when the compute module is swapped. End-of-life review of Payload Scale Interface found none identified. Test coverage on Payload Scale Interface is 36 per cent line coverage and the measurement basis was confirmed with Ines Duarte. On Payload Scale Interface the conclusion is qualified: the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. Payload Scale Interface publishes a stable interface, though two fields used by the gripper controller and the order execution path are documented only in code comments. Payload Scale Interface has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Payload Scale Interface is owned by the Robot Firmware team, with Ines Duarte named as the engineering owner of record. For Payload Scale Interface the reviewer worked from the interface specification, calibration records for nine units and a module-swap test. In production Payload Scale Interface is edge, only on robots fitted with the weighing deck, unchanged since the last architecture review. Hosting for Payload Scale Interface is on-robot Linux, user-space driver, and its C++17 code writes through to a local calibration record. The objective recorded against Payload Scale Interface is that payload mass is reported within 2 per cent of the reference load, and it is reported weekly to the engineering leadership. Payload Scale Interface rests on a single source, and the reviewer has said so rather than implying corroboration.

Schedule Q - Pipeline stage duration, failure rate and gate configuration (TDR-2026-10)

Pipeline stage	Mean duration	p90 duration	Failure rate	Flaky rate	Gate	Owning team
lint and format	1 m 10 s	1 m 40 s	3.1%	0.1%	blocking	Developer Productivity
unit tests, cloud services	6 m 20 s	9 m 05 s	5.4%	1.9%	blocking	Fleet Services
unit tests, robot software	11 m 45 s	18 m 30 s	7.8%	4.6%	blocking	Autonomy Core
static analysis, robot software	14 m 05 s	22 m 10 s	6.2%	0.4%	blocking	Autonomy Core
build, cloud container images	8 m 15 s	12 m 40 s	2.7%	0.3%	blocking	Platform Infrastructure
build, robot firmware images	19 m 30 s	31 m 20 s	4.1%	0.6%	blocking	Robot Firmware
integration tests, cloud	24 m 50 s	41 m 15 s	12.6%	8.2%	blocking	Fleet Services
contract tests, warehouse connectors	9 m 40 s	15 m 55 s	9.3%	3.7%	blocking	Integrations

4. The services carrying the largest debt - part 2 of 2

The 4. The services carrying the largest debt begun above continues from this point, and its row references run consecutively across the parts rather than restarting. The parts are to be read together.

Whatever caveat the 4. The services carrying the largest debt carried at its head travels with these rows, and no additional caveat attaches to this part. The sequence is unbroken either side of it.

Schedule Q - Pipeline stage duration, failure rate and gate configuration (TDR-2026-10) (continued, part 2 of 2)

Pipeline stage	Mean duration	p90 duration	Failure rate	Flaky rate	Gate	Owning team
simulation suite, short	33 m 10 s	52 m 00 s	8.9%	5.1%	blocking	Autonomy Core
simulation suite, nightly	3 h 40 m	4 h 25 m	14.2%	6.8%	advisory	Autonomy Core
hardware-in-the-loop, bench	1 h 15 m	2 h 05 m	10.4%	2.2%	blocking for firmware	Robot Firmware
dependency and licence scan	4 m 25 s	6 m 50 s	1.8%	0.2%	advisory	Security Engineering
secret scan	2 m 05 s	3 m 15 s	0.9%	0.1%	blocking	Security Engineering
container image vulnerability scan	5 m 35 s	8 m 20 s	16.7%	1.1%	advisory	Security Engineering
performance smoke, cloud	12 m 20 s	19 m 45 s	6.6%	3.4%	advisory	Platform Infrastructure
release packaging and signing	7 m 50 s	11 m 10 s	2.2%	0.2%	blocking	Developer Productivity

5. Debt with a capacity consequence

The two sites, shared region test, recorded as scenario LT-2025-14, was executed on 2025-10-14. The point of two sites, shared region run LT-2025-14 within the series is whether control and telemetry shared a topic. No limit reached was the constraint run LT-2025-14 exposed. load run LT-2025-14 has no single accountable owner, and that is the finding the reviewer would press hardest.

Reading load run LT-2025-15 for the binding constraint gives traffic manager Redis connection pool saturated at 84 per cent. 1,120 ms at p95 is the figure that matters in four sites, shared region run LT-2025-15, against 355 ms median

and 2,240 ms at the tail. The environment behind load run LT-2025-15 carries a synthetic robot fleet driven from recorded mission traces, and the reviewer weighted the result accordingly. LT-2025-15 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The percentile set for scenario LT-2025-16 reads 470 ms, 1,680 ms and 3,410 ms, with errors at 0.34%. Scenario LT-2025-16 is best read against the neighbouring run in the same series, from which it differs in whether the read replicas were enabled. On 2025-10-17 the team ran scenario LT-2025-16, the six sites, shared region case, at 720 robots. load run LT-2025-16 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Scenario LT-2025-17 used the standard performance environment, which is characterised by production data volumes and a stubbed paging provider. LT-2025-17 sits against the fifteen-minute map publication objective; whether that target is the right one is a separate question. Errors in load run LT-2025-17 were 1.24%, with latency of 620 ms, 2,350 ms and 5,900 ms across the three percentiles. order burst, ten times nominal for two minutes run LT-2025-17 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Applying the fifteen-minute map publication objective to telemetry burst with control traffic on the shared topic run LT-2025-18 produces the verdict recorded above. Load run LT-2025-18 exercised the telemetry burst with control traffic on the shared topic case with 480 robots and 9,400 orders an hour, on 2025-10-22. What limited scenario LT-2025-18 was reproduced the November 2025 control delay before the topic split. telemetry burst with control traffic on the shared topic run LT-2025-18 is the clearest instance in TDR-2026-10 of engineering practice being better than the documentation suggests.

Reconnect storm from the edge agents on failover was the constraint scenario LT-2025-19 exposed. 480 robots and 9,400 orders an hour define load run LT-2025-19, run on 2025-10-23. Outcome of run LT-2025-19 was fail against the error rate target, and the reviewer read the raw result rather than the summary. run LT-2025-19 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Configuration of the environment for database failover, allocator primary run LT-2025-20 included production data volumes and a stubbed paging provider. Against the stated targets, load run LT-2025-20 is pass with observation. Configuration for database failover, allocator primary run LT-2025-20 was 240 robots against 4,900 orders an hour in the database failover, allocator primary case. load run LT-2025-20 would be a candidate for the first post-close engineering review under any reasonable plan.

Reading LT-2026-01 for the binding constraint gives the control topic split removed the telemetry coupling. LT-2026-01 used the standard performance environment, which is characterised by a fleet simulator rather than physical robots, with the control loop unchanged. Configuration for scenario LT-2026-01 was 720 robots against 13,800 orders an hour in the six sites after the topic split case. load run LT-2026-01 is reported at the evidence strength available and has not been promoted to fit the narrative.

960-robot run LT-2026-02 supports this conclusion: the headroom claim is supportable at the tested scale and not at the scale in the operating plan. Latency in load run LT-2026-02 was 455 ms at p50, 1,340 ms at p95 and 2,880 ms at p99, with an error rate of 0.27%. In run LT-2026-02 the bottleneck was allocator single-writer database remains the limit. load run LT-2026-02 is reported at the evidence strength available and has not been promoted to fit the narrative.

In load run LT-2026-03 the median was 400 ms and the tail reached 2,310 ms; errors ran at 0.21%. LT-2026-03 shows where the system runs out of headroom: read offload bought headroom without a schema change. Before reading load run LT-2026-03, note that the environment carries the current production topology with one region held out. scenario LT-2026-03 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Outcome of scenario LT-2026-04 was fail against the 1,200 ms p95 target, and the reviewer read the raw result rather than the summary. Scenario LT-2026-04 shows where the system runs out of headroom: the allocator write path needs partitioning to go further. The 1,200 ms p95 objective published for order assignment is the target applied to LT-2026-04. scenario LT-2026-04 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

The verdict on 480-robot run LT-2026-05 is pass with observation. Marginally over the p95 target under burst was the constraint order burst after the broker queue change run LT-2026-05 exposed. The order burst after the broker queue

change test, recorded as run LT-2026-05, was executed on 2026-02-18. 480-robot run LT-2026-05 is the clearest instance in TDR-2026-10 of a control that exists on paper and is weaker in practice.

In the series scenario LT-2026-06 is the run that changes whether the read replicas were enabled. Run LT-2026-06 returned 430 ms median, 1,280 ms at the 95th percentile and 3,120 ms at the 99th, erroring at 0.36%. LT-2026-06 supports this conclusion: the result is consistent with the incident record, which is the strongest corroboration available. load run LT-2026-06 rests on a single source, and the reviewer has said so rather than implying corroboration.

Schedule P - Load-test matrix against the published latency targets (TDR-2026-10)

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
LT-2025-11	single site, steady state	120	1,850	210 ms	480 ms	760 ms	0.02%	pass
LT-2025-12	single site, peak hour	120	2,640	265 ms	690 ms	1,240 ms	0.05%	pass
LT-2025-13	single site, peak with a charger outage	120	2,640	310 ms	870 ms	1,690 ms	0.11%	pass with observation
LT-2025-14	two sites, shared region	240	4,900	290 ms	820 ms	1,510 ms	0.09%	pass
LT-2025-15	four sites, shared region	480	9,400	355 ms	1,120 ms	2,240 ms	0.18%	pass with observation
LT-2025-16	six sites, shared region	720	13,800	470 ms	1,680 ms	3,410 ms	0.34%	fail against the 1,200 ms p95 target
LT-2025-17	order burst, ten times nominal for two minutes	240	12,000	620 ms	2,350 ms	5,900 ms	1.24%	fail against the 1,200 ms p95 target
LT-2025-18	telemetry burst with control traffic on the shared topic	480	9,400	540 ms	4,100 ms	11,300 ms	0.42%	fail
LT-2025-19	region failover during peak	480	9,400	410 ms	1,950 ms	6,700 ms	2.85%	fail against the error rate target
LT-2025-20	database failover, allocator primary	240	4,900	340 ms	1,410 ms	4,200 ms	0.96%	pass with observation
LT-2026-01	six sites after the topic split	720	13,800	395 ms	1,090 ms	2,050 ms	0.14%	pass
LT-2026-02	eight sites, shared region	960	18,200	455 ms	1,340 ms	2,880 ms	0.27%	fail against the 1,200 ms p95 target
LT-2026-03	eight sites with the allocator read replicas enabled	960	18,200	400 ms	1,150 ms	2,310 ms	0.21%	pass
LT-2026-04	ten sites, shared region	1200	22,600	520 ms	1,720 ms	4,050 ms	0.48%	fail against the 1,200 ms p95 target
LT-2026-05	order burst after the broker queue change	480	24,000	480 ms	1,240 ms	2,760 ms	0.31%	pass with observation
LT-2026-06	shard rebalance under peak load	720	13,800	430 ms	1,280 ms	3,120 ms	0.36%	pass with observation
LT-2026-07	identity service token storm	960	18,200	180 ms	520 ms	1,010 ms	0.03%	pass
LT-2026-08	console session concurrency	480	9,400	290 ms	740 ms	1,480 ms	0.07%	pass

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
LT-2026-09	console session concurrency at twice the plan	480	9,400	380 ms	1,460 ms	3,900 ms	0.52%	fail
LT-2026-10	warehouse connector burst, one tenant	240	8,800	610 ms	1,830 ms	4,400 ms	0.68%	fail against the 1,500 ms p95 acknowledgement target
LT-2026-11	warehouse connector burst after outbox partitioning	240	8,800	440 ms	1,210 ms	2,540 ms	0.29%	pass
LT-2026-12	telemetry ingest at twice the current fleet	2400	n/a	1.4 s to console	22 s	48 s	0.08%	pass
LT-2026-13	twelve sites, projected 2027 fleet	1440	27,000	690 ms	2,450 ms	6,100 ms	0.74%	fail
LT-2026-14	four hour edge buffer drain after an outage	720	13,800	not applicable	not applicable	not applicable	0.00%	pass with observation
LT-2026-15	map publication to a thousand robots	1000	n/a	3 m 20 s	8 m 40 s	13 m 10 s	0.10%	pass
LT-2026-16	config distribution with thirty per cent of robots offline	720	13,800	1 m 50 s	4 m 30 s	9 m 20 s	0.00%	pass with observation

6. Where debt produced an incident

Assessed on its own record, tooling incident INC-2026-021 shows that management's severity rating is defensible and its remediation priority is not. A contributing factor in Simulation Harness incident INC-2026-021 was that the build pinned the compiler but not the optimisation flags used for release builds. Evidence strength on the tooling event INC-2026-021 analysis is strong on the cause and limited on the contributing factor. On delivery, internal incident INC-2026-021 shows this: not landed; the determinism check is written but not enabled. The blast radius of internal incident INC-2026-021 was no production site and no robot directly, the exposure being to internal systems. internal incident INC-2026-021 is reported at the evidence strength available and has not been promoted to fit the narrative.

The detection path for tooling event INC-2026-021 was the autonomy team noticed a step change in the pass rate. What went wrong in incident INC-2026-021 was that the physics core is not deterministic across optimisation levels. The immediate event in INC-2026-021 was that a nightly simulation suite produced non-reproducible results after the physics core was rebuilt, dated 2026-02-24. In taxonomy terms tooling incident INC-2026-021 is a tooling incident in Simulation Harness. Remediation for INC-2026-021 is to pin the full build configuration and add a determinism check to the harness; Colin Achebe owns it and the committed date is 2026-05-31. Simulation Harness incident INC-2026-021 would move from a concern to a verified defect if one further measurement were supplied.

The affected population in INC-2026-009 was one site and 4 robots. 1 h 30 m to quarantine the affected robots elapsed before calibration incident INC-2026-009 was mitigated, with resolution at 48 days. The judgement recorded against Blue Mesa Components incident INC-2026-009 is that an operational workaround masked the defect for longer than the incident record suggests. On delivery, calibration event INC-2026-009 shows this: landed on 2026-03-16 with a workflow gate. On INC-2026-009 the evidence is moderate, resting on the team's own post-mortem and one supporting log extract; the severity assessment is stated separately below. calibration incident INC-2026-009 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

On 2026-01-27 the record for calibration event INC-2026-009 opens: a gripper applied more force than the configured limit after a mechanical replacement without recalibration. A contributing factor in calibration event INC-2026-009 was that the service workflow allowed the step to be marked complete without a calibration record. In taxonomy terms INC-2026-009 is a calibration incident in Gripper Controller. How calibration incident INC-2026-009 surfaced matters to the assessment: the payload damage was reported by the customer. Calibration event INC-2026-009 was caused by a design gap: force limiting depends on a calibration that is only triggered by a service workflow step. INC-2026-009 was one of two items where TDR-2026-10 could not obtain the underlying data within the engagement.

Data integrity incident INC-2025-058 was raised on 2025-05-09 after two tenants reused the same order identifier namespace and duplicate suppression discarded live orders. The contributing factor recorded against data integrity incident INC-2025-058 is that the integration specification allowed the identifier to be tenant-scoped but did not require it. Detection came through the customer reported missing picks the same afternoon, which places INC-2025-058 on the reported side of the ledger. Data integrity event INC-2025-058 traces back to a single fact: duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier. The reviewer's reading of incident INC-2025-058 is that the failure was in the detection design rather than in the code that failed. incident INC-2025-058 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Grading INC-2025-058 on evidence rather than on consequence gives limited, being a single account corroborated only by the ticket timeline. The reviewer checked the INC-2025-058 commitment against the code and release records; landed on 2025-05-30; the backfill was completed on 2025-06-03. The remediation committed after INC-2025-058 was to key suppression on the tenant and identifier pair and backfill the affected order records, owned by Tomas Villareal with a due date of 2025-07-15. The affected population in incident INC-2025-058 was 2 site or tenant with no robot taken out of service. The classification applied to Dunbar Industrial incident INC-2025-058 is data integrity, and the affected component is WMS Connector. data integrity incident INC-2025-058 appears twice in the working papers for TDR-2026-10 because the first answer was corrected after a follow-up request.

The reviewer treats one contributing factor in scheduling incident INC-2025-052 as the more important lesson: the site had changed its shift pattern without notifying the deployment team. Scheduling event INC-2025-052 belongs to the scheduling group and sits with Charge Scheduler. Following incident INC-2025-052 the team undertook to release charge windows on a shift-end event and reconcile the calendar nightly, with Nadia Semenova accountable by 2025-06-30. Time to mitigate scheduling incident INC-2025-052 was 4 h 10 m, and time to resolve was 16 days. The reviewer checked the incident INC-2025-052 commitment against the code and release records; landed on 2025-06-11 with a nightly reconciliation report. INC-2025-052 has no single accountable owner, and that is the finding the reviewer would press hardest.

Root cause in incident INC-2025-052 was that the scheduler treated a shift calendar entry as immutable once the window had opened. On 2025-04-21 the record for incident INC-2025-052 opens: an early shift end left opportunity-charge windows reserved and the following shift started with a depleted fleet. Drawing the evidence in Charge Scheduler incident INC-2025-052 together, the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. The detection path for Evergreen Warehousing incident INC-2025-052 was the site's own shift report, escalated the next morning. Charge Scheduler incident INC-2025-052 touched one site and 44 robots, and Evergreen Warehousing was the affected account. scheduling incident INC-2025-052 was checked against the corresponding entry in the company's own register, and the register was current.

Incident INC-2025-110 affected 1 site or tenant with no robot taken out of service. Incident INC-2025-110 began on 2025-09-03 when a late telemetry partition was excluded from the daily load and a customer report understated throughput. The remediation committed after Analytics Warehouse Loader incident INC-2025-110 was to fail the load on an incomplete partition and reprocess the affected days, owned by Farida Cisse with a due date of 2025-10-31. Data integrity incident INC-2025-110 would have been shorter or smaller had it not been that the partition completeness check was written but never wired into the load gate. Data integrity incident INC-2025-110 was detected because the customer challenged the report three days later. INC-2025-110 would be a candidate for the first post-close engineering review under any reasonable plan.

Drawing the evidence in incident INC-2025-110 together, the team's analysis was honest about a gap it could have concealed. Time to mitigate data integrity event INC-2025-110 was not applicable; reporting only, and time to resolve

was 18 days. The engineering team's root cause for incident INC-2025-110 is that the loader treated a missing partition as empty rather than as incomplete, and the reviewer found the analysis sound. Strong on the cause and limited on the contributing factor is the evidence grade for Analytics Warehouse Loader incident INC-2025-110, and the reader should not read severity into it. Data integrity is the class of data integrity event INC-2025-110; Analytics Warehouse Loader is the component. incident INC-2025-110 is reported at the evidence strength available and has not been promoted to fit the narrative.

The trigger recorded for incident INC-2025-071 on 2025-06-11 was that a regional ingest endpoint failed over and edge buffers filled during a fifty-two minute outage. Contributing to incident INC-2025-071, and not repaired by the remediation, is that the upgrade runbook did not require a partition leadership check between broker restarts. The corrective action for availability incident INC-2025-071 is to add a leadership and under-replicated partition check to the broker upgrade runbook, and 2025-07-31 is the date the team gave. Drawing the evidence in incident INC-2025-071 together, the blast radius was set by a deployment topology decision made years earlier. Availability incident INC-2025-071 took 52 m to stabilise and 52 m to close out. availability event INC-2025-071 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

INC-2025-071 was contained to 6 sites and 214 robots. Telemetry Ingest incident INC-2025-071 is recorded as follows on delivery: landed on 2025-07-18 and exercised in the September failover drill. Evidence strength on the multiple sites incident INC-2025-071 analysis is limited, being a single account corroborated only by the ticket timeline. Attention came to incident INC-2025-071 when the ingest availability alert fired within ninety seconds. Telemetry Ingest accounts for availability event INC-2025-071, a availability event. availability incident INC-2025-071 is treated as material to the investment case rather than as a routine engineering item.

The corrective action for multiple sites incident INC-2025-124 is to annotate staleness on every read path and add a rebalance-under-load test to the platform suite, and 2025-11-30 is the date the team gave. Consistency event INC-2025-124 was mitigated in 1 h 08 m; the resolution ran to 29 days. Incident INC-2025-124 was raised on 2025-10-01 after a shard rebalance served shadow state up to ninety seconds stale without marking it stale. Grading consistency event INC-2025-124 on evidence rather than on consequence gives moderate; the narrative is consistent but the reviewer could not independently reproduce it. In taxonomy terms Robot Shadow Store incident INC-2025-124 is a consistency incident in Robot Shadow Store. multiple sites incident INC-2025-124 rests on a single source, and the reviewer has said so rather than implying corroboration.

Consistency event INC-2025-124 carries a second-order finding: no test exercised a shard rebalance while the cluster was under load. The affected population in incident INC-2025-124 was 5 sites and 178 robots. Multiple sites incident INC-2025-124 reached the engineering team when the console latency alert fired, though the underlying cause took a day to attribute. The judgement recorded against incident INC-2025-124 is that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. Consistency event INC-2025-124 is recorded as follows on delivery: landed on 2025-10-30; the load test was added on 2025-11-12. consistency event INC-2025-124 would be a candidate for the first post-close engineering review under any reasonable plan.

On Fleet Health Monitor incident INC-2026-028 the evidence is strong, resting on the production traces the team supplied; the severity assessment is stated separately below. Observability incident INC-2026-028 began on 2026-03-10 when two distinct faults at one site were grouped into a single case and the second went unworked for two days. The engineering team's root cause for observability incident INC-2026-028 is that alert grouping keyed on site and severity rather than on the underlying signal, and the reviewer found the analysis sound. Detection came through the customer asked why one symptom persisted after the case had been closed, which places Fleet Health Monitor incident INC-2026-028 on the reported side of the ledger. Observability event INC-2026-028 classifies as a observability event against Fleet Health Monitor. Fleet Health Monitor incident INC-2026-028 would move from a concern to a verified defect if one further measurement were supplied.

Verification of the observability incident INC-2026-028 remediation gives landed on 2026-04-10. The reviewer's reading of observability incident INC-2026-028 is that the team's analysis was honest about a gap it could have concealed. The reviewer treats one contributing factor in incident INC-2026-028 as the more important lesson: case closure did not check for unresolved constituent alerts. Wanda Njoku committed after observability event INC-2026-028 to group on the signal and block closure while a constituent alert remains active, due 2026-04-30. Not a service outage; 2

days of delayed repair elapsed before Fleet Health Monitor incident INC-2026-028 was mitigated, with resolution at 31 days. observability event INC-2026-028 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

7. Debt with a security consequence

SEC-2025-03 against Remote Operations Console is supported by the fact that demonstrated in the tester's laboratory network, not on a customer site; evidence strength is limited, resting on a configuration read rather than on a demonstrated exploitation. Remote Operations Console finding SEC-2025-03 translates into the following exposure: an on-path attacker on a customer network could observe operator commands. SEC-2025-03 against Remote Operations Console is remediated, with a closure date of 2026-01-22. Finding SEC-2025-03 is assessed as follows: management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. SEC-2025-03 against Remote Operations Console is the clearest instance in TDR-2026-10 of engineering practice being better than the documentation suggests.

Remediation status on assessment finding SEC-2026-13 is remediated; the record shows 2026-05-15. Order Broker finding SEC-2026-13 is rated Low with a numeric band of 3.3 on the tester's scoring scale. A large payload degrades broker throughput is the exposure SEC-2026-13 against Order Broker describes. Substantiating SEC-2026-13 is the fact that throughput degradation measured by the tester, graded moderate. Order Broker finding SEC-2026-13 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

After the fix, SEC-2026-11 against Gripper Controller carries residual risk described as medium; the safety controller limit is the compensating control. As at the review date Gripper Controller finding SEC-2026-11 was open and the closure entry reads committed 2026-11-30. Finding SEC-2026-11 matters because a non-privileged process could alter a force limit parameter. On assessment finding SEC-2026-11 the tester's evidence was that file permissions observed on a laboratory robot, and the reviewer grades it strong. Gripper Controller finding SEC-2026-11 is the clearest instance in TDR-2026-10 of engineering practice being better than the documentation suggests.

After the fix, assessment finding SEC-2026-08 carries residual risk described as low; role grants now require a second approver and are alerted. The evidence for SEC-2026-08 is moderate: role grant performed in a staging tenant. The tester scored assessment finding SEC-2026-08 at 6.4 and labelled it Medium. The finding at finding SEC-2026-08, dated 2026-04-18, is that identity service admin role assignable without a second approval. SEC-2026-08 is included because a reader of TDR-2026-10 should be able to disagree with the inference without repeating the work.

The finding at assessment finding SEC-2026-07, dated 2026-04-17, is that hardware-in-the-loop rig controller runs an unsupported operating system. The remediation ledger shows Medium-severity finding SEC-2026-07 as open, committed 2026-10-31. SEC-2026-07 against Hardware-in-the-Loop Rig arrives as a Medium finding with a 6.7 score, and the reviewer did not adjust either. An unpatched laboratory host sits on the engineering network is the exposure Medium-severity finding SEC-2026-07 describes. Medium-severity finding SEC-2026-07 was excluded from an earlier draft of TDR-2026-10 and reinstated after the reviewer obtained the supporting record.

Impact of SEC-2025-07 is not theoretical; the same secret was reusable at any site provisioned from that template. Severity on assessment finding SEC-2025-07 is Medium, scored 5.9 by the assessor. Weighing finding SEC-2025-07 against the compensating controls, the underlying pattern appears in two other components that the tester did not reach. Remediation status on assessment finding SEC-2025-07 is remediated; the record shows 2026-01-16. assessment finding SEC-2025-07 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Rating for SEC-2025-09 is Medium, band 6.3; the reviewer re-read the scoring rationale and accepted it. Read alongside the remediation record, SEC-2025-09 shows that the remediation is adequate and the time to remediate is not. The residual position on SEC-2025-09 is medium; a two-person rule is designed but not enforced in the tool, stated after mitigation rather than assuming full success. Tracking assessment finding SEC-2025-09 to closure gives a status of partially remediated and a date of pending. assessment finding SEC-2025-09 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Finding SEC-2026-18 is not fully retired by its remediation: low; no fleet impact demonstrated. Assessment finding SEC-2026-18 is assessed as follows: the fix is verified and the class of defect is not, since no test now prevents its return. Finding SEC-2026-18 sits against Battery Management Service and records that battery pack diagnostic interface has no

rate limit. The tester scored Low-severity finding SEC-2026-18 at 3.4 and labelled it Low. finding SEC-2026-18 has no single accountable owner, and that is the finding the reviewer would press hardest.

Costmap Fusion finding SEC-2026-16 is rated Medium with a numeric band of 5.8 on the tester's scoring scale. For Medium-severity finding SEC-2026-16 the record shows that connected from a second host on the robot network, which is strong evidence and is graded separately from severity. Costmap fusion diagnostic socket bound to all interfaces is the substance of finding SEC-2026-16, raised on 2026-04-22. Tracking Costmap Fusion finding SEC-2026-16 to closure gives a status of remediated and a date of 2026-06-26. SEC-2026-16 would move from a concern to a verified defect if one further measurement were supplied.

Finding SEC-2026-14 is assessed as follows: the risk acceptance lacks an expiry date, which is a process finding in its own right. Traffic manager admin action lacked an audit record is the substance of SEC-2026-14, raised on 2026-04-21. SEC-2026-14 against Traffic Manager is not fully retired by its remediation: low. The evidence for SEC-2026-14 against Traffic Manager is limited, resting on a configuration read rather than on a demonstrated exploitation: action performed and the audit trail checked. Traffic Manager finding SEC-2026-14 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

On assessment finding SEC-2026-15 the conclusion is that the remediation is adequate and the time to remediate is not. Residual risk after remediation of Low-severity finding SEC-2026-15 is low; management judged the operational cost of enforcement higher than the risk. The reason Low-severity finding SEC-2026-15 was not downgraded is that operator confusion rather than a direct security impact. Low is the rating finding SEC-2026-15 carries, at 3.0 on the assessor's own scale. SEC-2026-15 against Site Provisioning Service has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

The finding at SEC-2025-05, dated 2025-09-19, is that telemetry ingest accepted a robot identity without checking device attestation. SEC-2025-05 stands remediated; 2026-02-06 is what the record gives for closure. After the fix, SEC-2025-05 carries residual risk described as low; attestation is now required on the ingest path. Rating for SEC-2025-05 is High, band 7.8; the reviewer re-read the scoring rationale and accepted it. assessment finding SEC-2025-05 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

What assessment finding SEC-2026-01 enables is this: one insider can move an unreviewed artefact to production. Read alongside the remediation record, SEC-2026-01 against Release Orchestrator shows that the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. On 2026-04-14 the tester recorded Release Orchestrator finding SEC-2026-01: emergency release path permits a single approver. Residual risk after remediation of assessment finding SEC-2026-01 is high until the second-approver control lands. assessment finding SEC-2026-01 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

8. Debt with a safety consequence

Register entry HZ-17 scores No direct harm on severity; exposure is Frequent and controllability Low. Reading Frequent-exposure hazard HZ-17 against the verification evidence, extraction runs weekly, so a busy robot can still lose events between extractions. The design measure recorded at No direct harm hazard HZ-17 is periodic fault log extraction to the cloud with a retention policy. hazard HZ-17 was checked against the company's own register, which had not been updated since the change landed.

The reviewer's note on Fire hazard HZ-11 is that state-of-health drift on opportunity-charged packs is an open modelling item. Field history for Improbable-exposure hazard HZ-11 is that it produced one field event that did not result in harm. Hazard HZ-11 remains Broadly acceptable, subject to the state-of-health drift item after the control is applied, and the register marks it partially verified. Improbable-exposure hazard HZ-11 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Reading HZ-03 against the verification evidence, robots without the weighing deck rely on operator loading discipline. Entry hazard HZ-03 in the hazard register reads: load falls from an elevated deck during transport. Status partially verified and residual risk Tolerable where the weighing deck is fitted are what the register gives for hazard HZ-03. Occasional-exposure hazard HZ-03 would be a candidate for the first post-close engineering review under any reasonable plan.

Evidence of verification at Minor injury hazard HZ-14 consists of console interaction test plus a review of session records. The parameter set for register entry HZ-14 reads Minor injury, Occasional, Low. Status verified and residual risk Tolerable once the stale roster defect is closed are what the register gives for hazard HZ-14 in the register. register entry HZ-14 is treated as material to the investment case rather than as a routine engineering item.

The parameter set for register entry HZ-21 reads Minor injury, Remote, Low. The residual entry for hazard HZ-21 reads Broadly acceptable, verified. Verification of register entry HZ-21 was by witnessed push test on two robot variants. Minor injury hazard HZ-21 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The reviewer's note on Serious injury hazard HZ-19 is that occupancy sensing is fitted at every dock revision currently deployed. HZ-19 was scoped to the two pallet-mover variants only, and the reviewer checked the scoping against the fleet record. Evidence of verification at Remote-exposure hazard HZ-19 consists of occupancy sensor bench test plus a witnessed dock approach. register entry HZ-19 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Mitigation for HZ-13 is detection of unmapped obstacles with a stop, plus a site procedure for barriers. Serious injury harm, Occasional exposure and Medium controllability are the parameters recorded for Serious injury hazard HZ-13. Exposure under hazard HZ-13 in the register extends to every site with a mezzanine or ramp. Occasional-exposure hazard HZ-13 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Tolerable at commissioned sites; two sites have unmapped corners is the residual position on Occasional-exposure hazard HZ-02, whose status is partially verified. Hazard HZ-02 in the register is the entry for the case that robot contacts a pedestrian emerging from a blind cross-aisle. Mitigation for hazard HZ-02 in the register is aisle-end speed reduction plus a mirrored sensing field at mapped blind corners. HZ-02 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Tolerable is the residual position on Remote-exposure hazard HZ-15, whose status is partially verified. The hazard recorded at Remote-exposure hazard HZ-15 is that robot obstructs an emergency egress route. Hazard HZ-15 in the register scores Serious injury on severity; exposure is Remote and controllability Medium. hazard HZ-15 in the register was excluded from an earlier draft of TDR-2026-10 and reinstated after the reviewer obtained the supporting record.

One caveat attaches to hazard HZ-16: the ring pause defect in July 2025 did not produce a mixed-version robot. The parameter set for HZ-16 reads Equipment damage, Remote, Low. Exposure under hazard HZ-16 in the register extends to the six sites with narrow-aisle geometry. register entry HZ-16 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The parameter set for HZ-09 reads Serious injury, Remote, Medium. Serious injury hazard HZ-09 was scoped to the sites that have moved to opportunity charging, and the reviewer checked the scoping against the fleet record. Hazard HZ-09 was verified through ramp testing on a wet and a dry surface at two sites. Remote-exposure hazard HZ-09 illustrates why TDR-2026-10 separates the severity of a finding from the confidence behind it.

The reviewer's note on HZ-12 is that one site's dock layout leaves less clearance than the layout guide requires. Field history for HZ-12 is that it produced two near-miss reports from site staff. Minor injury hazard HZ-12 carries residual risk of Tolerable with a status of verified. hazard HZ-12 would be a candidate for the first post-close engineering review under any reasonable plan.

9. Debt exposure by site

Ontario, California is notable in this review because it is the site whose shift pattern change caused the April 2025 charge incident. 2024-05 is the go-live month at Ontario, California, where Evergreen Warehousing now runs 88 robots. Ontario, California is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Meridian, Idaho is the longest-running customer deployment and the source of the most incident history, and the reviewer weighted the sample accordingly. Asterion Logistics operates 96 Helioscope robots at Meridian, Idaho,

commissioned in 2022-09. Meridian, Idaho is treated as material to the investment case rather than as a routine engineering item.

2021-06 is the go-live month at Fremont, California, where internal engineering site now runs 34 robots. The reason Fremont, California appears here is that it is the reference site, and the only one with a laboratory alongside. Fremont, California is one of the entries where TDR-2026-10 relies on the company's own measurement rather than an independent one.

For Grand Prairie, Texas the material fact is that it is the second Asterion site, commissioned on the same template. Headroom at Grand Prairie, Texas stands at 31%; vendor A is the integrated warehouse system. Grand Prairie, Texas was cross-checked against the engineering record supplied for TDR-2026-10, and the two agree.

At Bethlehem, Pennsylvania the fleet is 37 robots, and the site has been live since 2025-11. Bethlehem, Pennsylvania is sparse survey features, which is why relocalisation is slow here, which is why the reviewer sampled it. Bethlehem, Pennsylvania is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Edwardsville, Illinois is the newest Crestline site, still ramping, and the reviewer weighted the sample accordingly. At Edwardsville, Illinois the fleet is 24 robots, and the site has been live since 2026-03. Edwardsville, Illinois carries a follow-up request that was satisfied after the draft of TDR-2026-10 was circulated.

2024-02 is the go-live month at Lehigh Valley, Pennsylvania, where Crestline Fulfillment now runs 64 robots. Lehigh Valley, Pennsylvania is commissioned during the connector rework, which is why the reviewer sampled it. Lehigh Valley, Pennsylvania has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

What distinguishes Chino, California is that it is the go-live delayed by the forked provisioning template. At Chino, California the fleet is 33 robots, and the site has been live since 2026-01. Chino, California is the only entry in its group where the reviewer and the engineering team reached different conclusions.

On Kenosha, Wisconsin the record shows that it is the second opportunity-charging site. Capacity headroom at Kenosha, Wisconsin is 49%, and the warehouse system is vendor D. Kenosha, Wisconsin was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Whitestown, Indiana runs 44 robots for Juniper Bay Apparel and opened in 2025-09. For Whitestown, Indiana the material fact is that it is the site with the standing map publication workaround. Whitestown, Indiana was inspected in the repository rather than described from the ticket, which is why the account is specific.

For Reno, Nevada the material fact is that it is the smallest customer deployment. Reno, Nevada reports 57% headroom against its integration with vendor D. Reno, Nevada is one of the entries where TDR-2026-10 relies on the company's own measurement rather than an independent one.

Groveport, Ohio is notable in this review because it is opportunity charging only, which is where the pack estimator drift appeared. Remaining capacity at Groveport, Ohio is 44%, measured against the site's peak order profile. Groveport, Ohio was cross-checked against the engineering record supplied for TDR-2026-10, and the two agree.

Tracy, California reports 36% headroom against its integration with vendor B. 58 robots are deployed at Tracy, California, which went live in 2025-01. Tracy, California carries a follow-up request that was satisfied after the draft of TDR-2026-10 was circulated.

At Newnan, Georgia the fleet is 66 robots, and the site has been live since 2025-07. What distinguishes Newnan, Georgia is that it is narrow-aisle geometry, and the source of the August 2025 mutual-yield stall. Newnan, Georgia rests on a single source, and the reviewer has said so rather than implying corroboration.

10. Items the company has chosen not to address

3 declined items are carried in the present report, of which two lack any documented compensating control. This report treats a backlog label as evidence of intent and not as evidence of remediation, which is why several items remain open in this report that the company regards as handled. In TDR-2026-10 the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. TDR-2026-10 follows the reviewer's usual practice

of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. The report at TDR-2026-10 deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.

The open items carried forward from this report are set out below; a reader should treat the committed dates as intentions rather than facts. Open items in the present report are stated with what would close each one, so a post-close plan can be costed against them. Two positions in TDR-2026-10 are unresolved between the reviewer and the engineering team, and both are left unresolved on the page. One declinature in report TDR-2026-10 rests on an engineering judgement the reviewer shares; the others rest on prioritisation, which is a different kind of argument. The technical-debt register, end-of-life dependency exposure and remediation follow-through produced more evidence than the present report reproduces, and the working papers hold the remainder.

Where the report at TDR-2026-10 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. The evidence convention in the present report is that severity describes consequence and evidence strength describes confidence, and the two are never merged. Sampling in this report covered 19 of the 34 catalogued services and 5 of the 18 live sites. 5 sites contributed evidence to the present report, and one further site was in acceptance testing and was excluded. J. Ferreiro has kept the working papers behind this report in a form that a second reviewer could re-perform.

The report at TDR-2026-10 proposes 6 metrics for the post-close monitoring pack, each already produced by an existing system. Report TDR-2026-10 recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. Coverage of report TDR-2026-10 is deliberately uneven: the safety and control path was examined closely and the reporting path only at a high level. J. Ferreiro chose the sample for this report after reading the dependency map, so the selection is not proportionate to headcount or to code volume. Wrenfield Technical Advisory LLP raised each finding with the owning engineer before it was written up, and the responses are recorded.

11. Conclusions on the debt position

Whatever weight a reader gives the individual findings, this report concludes that the technical estate would support the business plan through the next twelve months and not through the fleet size the plan assumes for 2027 without the identified programme. For Ardent Peak's purposes, the present report concludes that the technical estate would support the business plan through the next twelve months and not through the fleet size the plan assumes for 2027 without the identified programme. TDR-2026-10 does not include a code audit; the reviewer read code where a specific claim required it and no further. The principal limitation on report TDR-2026-10 is time: 7 weeks was enough to test the control path and not enough to test the reporting path. J. Ferreiro has separated what the company has built from what the company can demonstrate about what it has built.

Report TDR-2026-10 recommends that 2 items be reflected in the disclosure schedules and that the remainder be tracked in the post-close plan. J. Ferreiro would put 5 items from TDR-2026-10 into the first hundred days rather than into a warranty. Nothing in report TDR-2026-10 is presented as verified on the strength of an interview alone. J. Ferreiro has left several concerns in the report at TDR-2026-10 at limited evidence strength rather than dropping them or dressing them up. J. Ferreiro has kept the company's terminology even where a different word would have read better, so that a reader can search the source documents.

The pack the present report proposes is deliberately short, because a longer one would be produced and not read. The monitoring pack TDR-2026-10 proposes would have surfaced two of the incidents examined here before a customer noticed. The present report did not rely on management representation alone; every material claim was traced to a document, a measurement or an observation. The approach J. Ferreiro took in this report was to test the company's own assertions rather than to restate them. Where the present report uses the word verified, it means the reviewer performed or witnessed something, and nothing weaker.